

TITLE:

Interactive Plotly/Dash dashboard

Dataset Used:

nhstalkingtherapies_month_mar_2026_activity_performance.csv

&

psych-ther-ann-rep-csv-2024-25-psych-meds-2

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DATA ANALYTICS INTERN

SUBMITTED TO:

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Datasets Introduction

IAPT Dataset Introduction

The IAPT dataset contains information related to NHS Talking Therapies services across England. It includes:

Main Information

- Referrals received
- Treatment activity
- Recovery rates
- Waiting times
- Completed treatments
- Regional service performance

Purpose of Dataset

The dataset is used to monitor:

- Mental health service access
- Patient outcomes
- Therapy effectiveness
- NHS regional performance

Example Fields

- Region
- Reporting Period
- Referrals
- Recovery Rate
- Waiting Time
- Finished Courses of Treatment

Prescribing Dataset Introduction

The prescribing dataset contains data about antidepressant and anxiolytic medication prescribing trends.

Main Information

- Prescription counts
- Medication categories
- Regional prescribing activity
- Monthly prescribing trends

Purpose of Dataset

The dataset helps analyze:

- Medication usage patterns
- Mental health prescribing growth
- Regional prescribing differences
- Healthcare demand trends

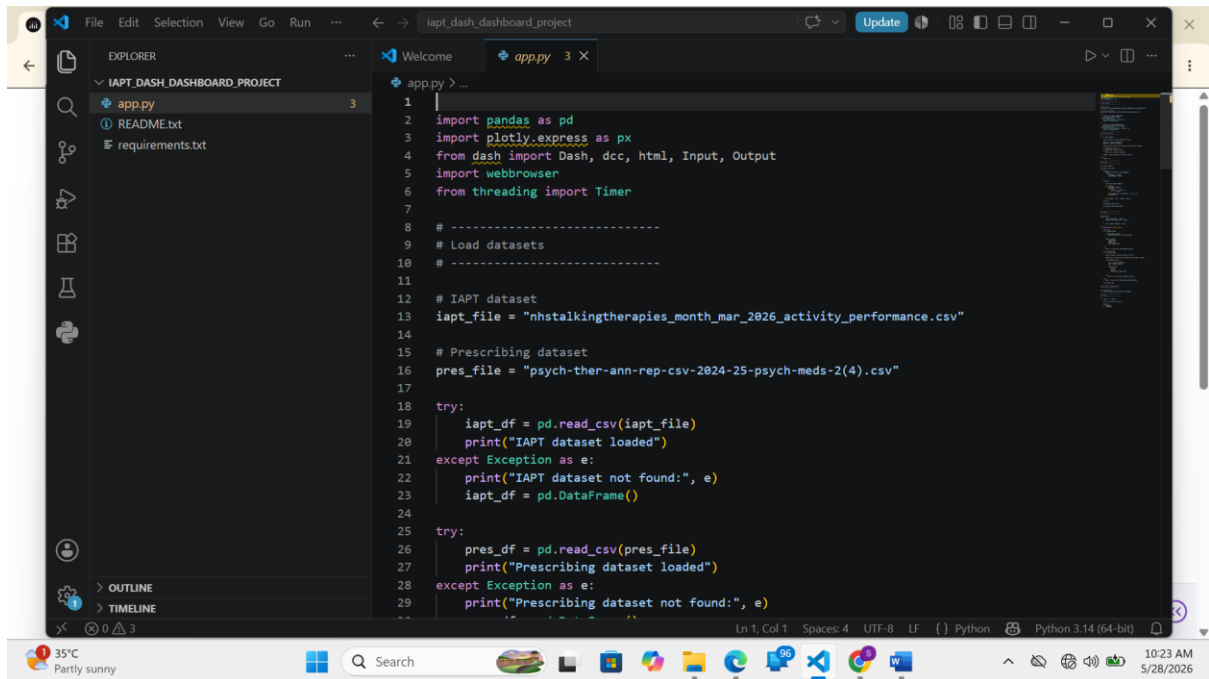
Example Fields

- Region
- Month
- Drug Category
- Prescription Count
- Total Items Prescribed

Technologies Used

Technology	Purpose
Python	Core programming language
Pandas	Data cleaning and preprocessing
Plotly	Interactive charts and visualizations
Dash	Web dashboard framework
Jupyter/VS Code	Development environment
NumPy	Numerical operations
HTML Components (Dash)	Dashboard layout
CSS Styling (Optional)	UI enhancement

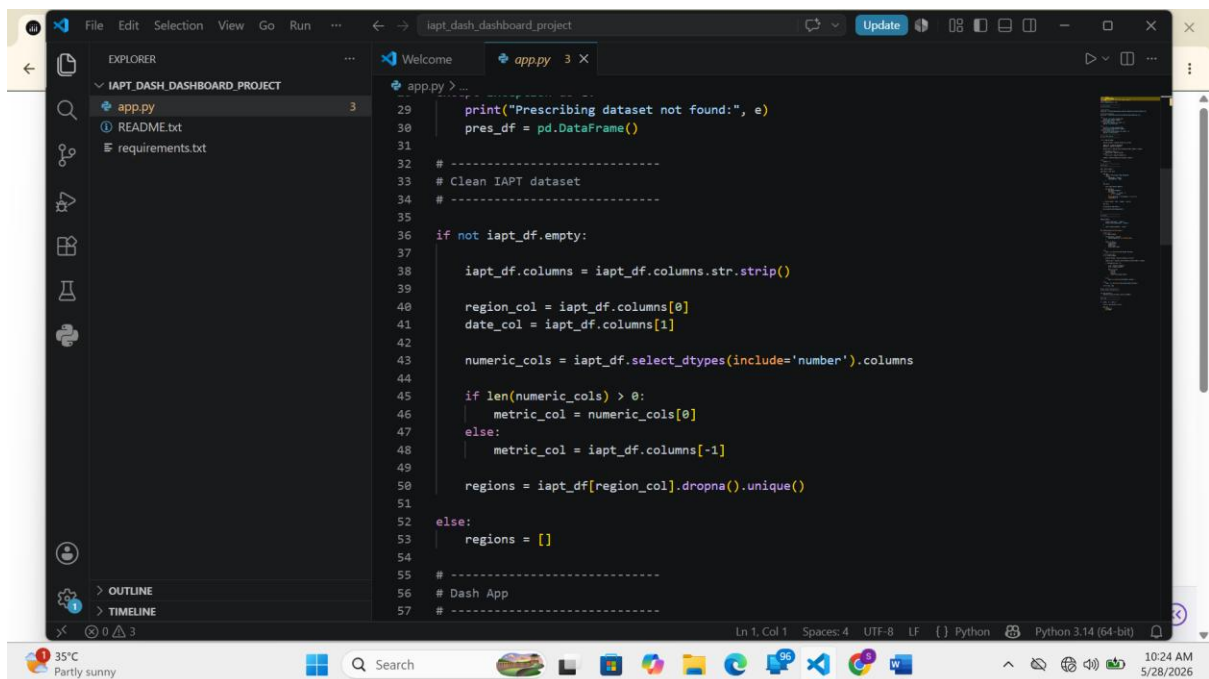
Interactive Plotly/Dash Dashboard



This screenshot shows the initial setup of the `app.py` file in the `IAPT_DASHBOARD_PROJECT`. The Explorer panel on the left lists `app.py`, `README.txt`, and `requirements.txt`. The main editor displays the following Python code:

```
1
2 import pandas as pd
3 import plotly.express as px
4 from dash import Dash, dcc, html, Input, Output
5 import webbrowser
6 from threading import Timer
7
8 # -----
9 # Load datasets
10 # -----
11
12 # IAPT dataset
13 iapt_file = "nhstalkingtherapies_month_mar_2026_activity_performance.csv"
14
15 # Prescribing dataset
16 pres_file = "psych-ther-ann-rep-csv-2024-25-psych-meds-2(4).csv"
17
18 try:
19     iapt_df = pd.read_csv(iapt_file)
20     print("IAPT dataset loaded")
21 except Exception as e:
22     print("IAPT dataset not found:", e)
23     iapt_df = pd.DataFrame()
24
25 try:
26     pres_df = pd.read_csv(pres_file)
27     print("Prescribing dataset loaded")
28 except Exception as e:
29     print("Prescribing dataset not found:", e)
```

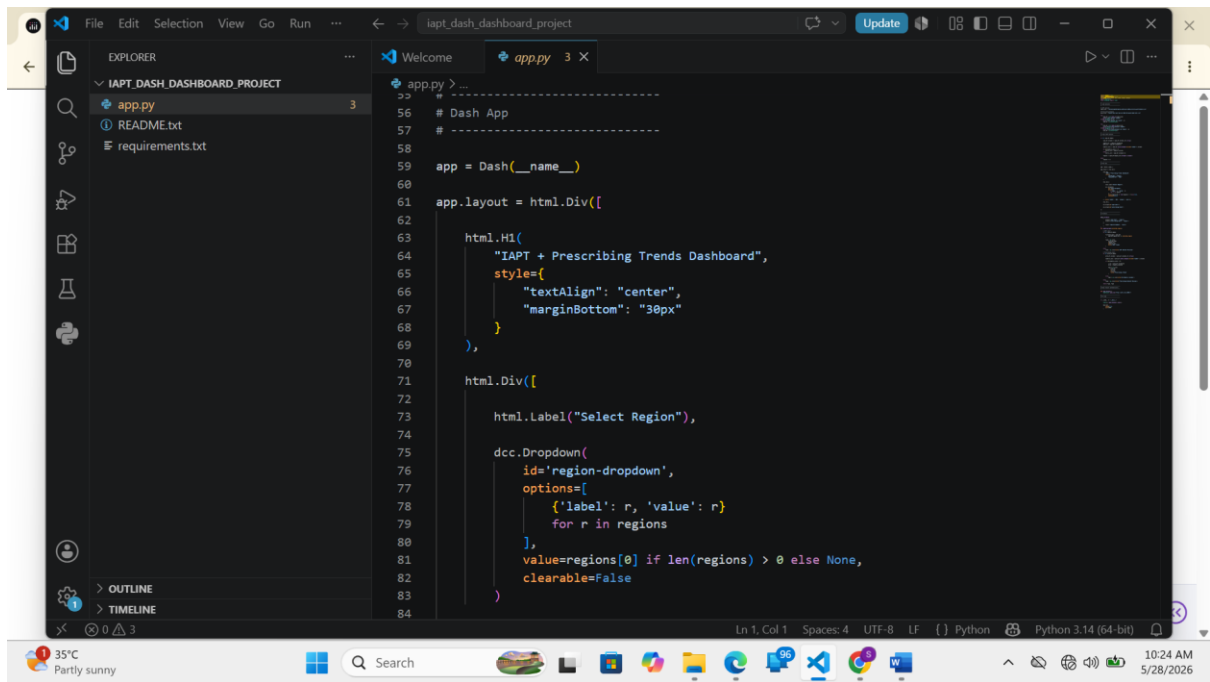
The status bar at the bottom indicates the file is at Line 1, Column 1, using 4 spaces for indentation, in UTF-8 encoding, with Python 3.14 (64-bit) installed.



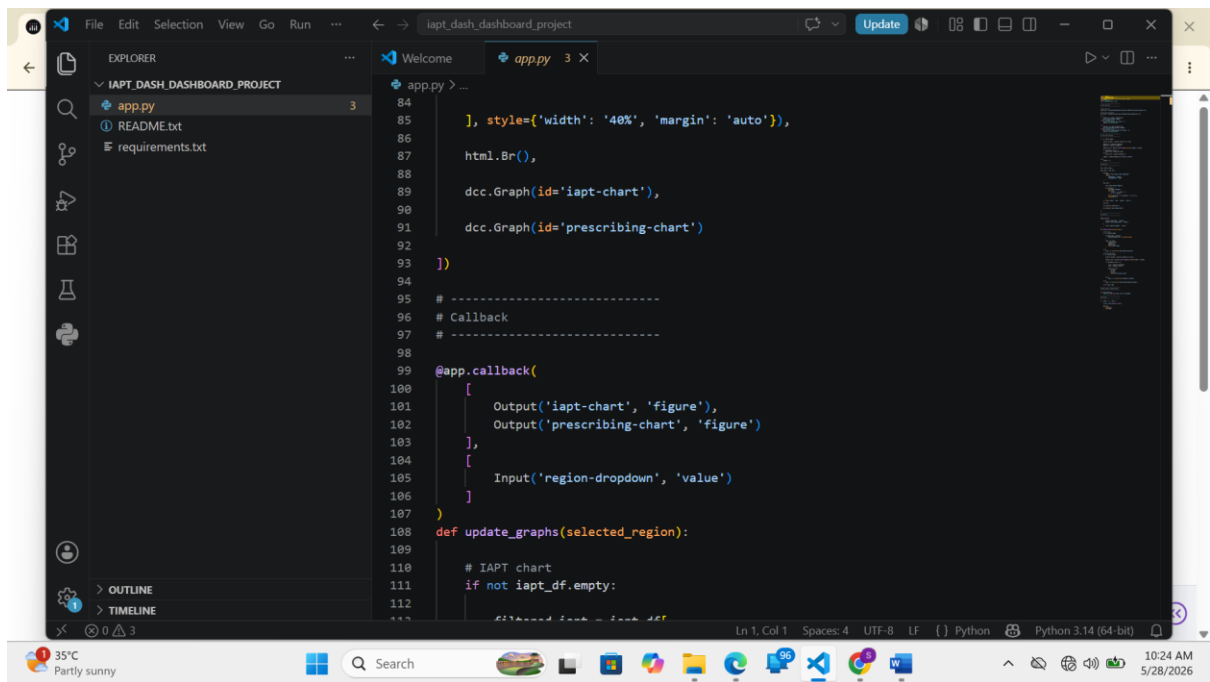
This screenshot shows the data processing logic in the `app.py` file. The Explorer panel on the left lists `app.py`, `README.txt`, and `requirements.txt`. The main editor displays the following Python code:

```
29 print("Prescribing dataset not found:", e)
30 pres_df = pd.DataFrame()
31
32 # -----
33 # Clean IAPT dataset
34 # -----
35
36 if not iapt_df.empty:
37     iapt_df.columns = iapt_df.columns.str.strip()
38
39     region_col = iapt_df.columns[0]
40     date_col = iapt_df.columns[1]
41
42     numeric_cols = iapt_df.select_dtypes(include='number').columns
43
44     if len(numeric_cols) > 0:
45         metric_col = numeric_cols[0]
46     else:
47         metric_col = iapt_df.columns[-1]
48
49     regions = iapt_df[region_col].dropna().unique()
50
51 else:
52     regions = []
53
54 # -----
55 # Dash App
56 # -----
57
```

The status bar at the bottom indicates the file is at Line 1, Column 1, using 4 spaces for indentation, in UTF-8 encoding, with Python 3.14 (64-bit) installed.



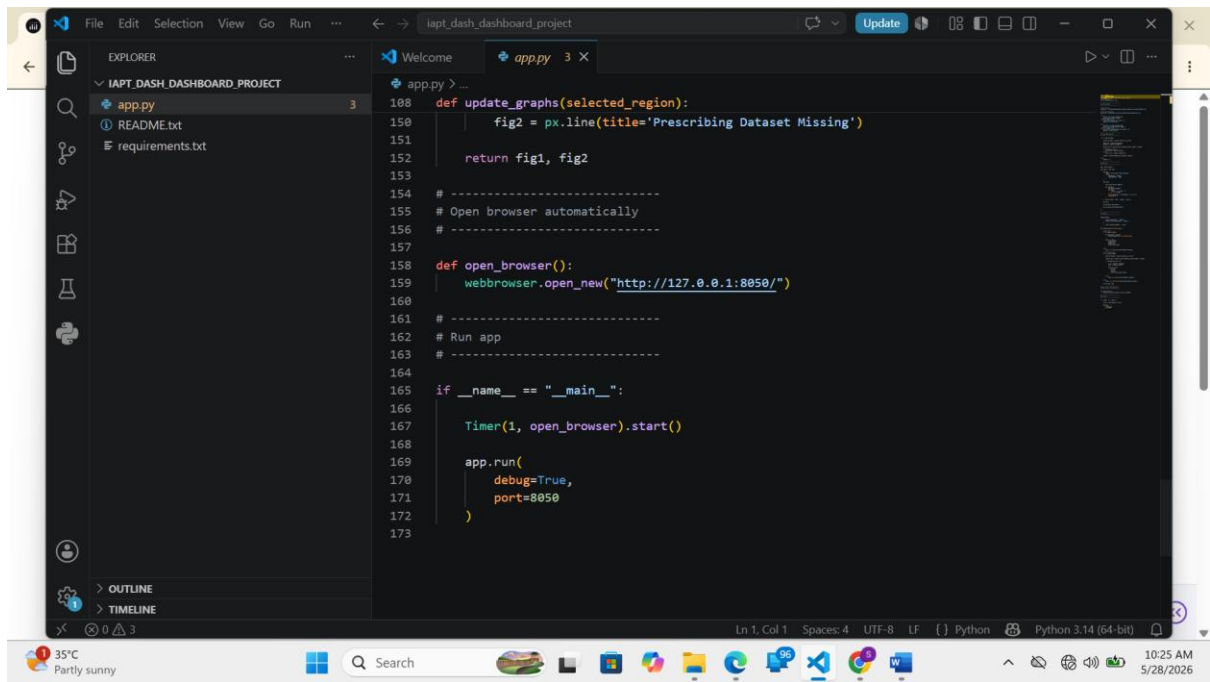
```
56 # Dash App
57 # -----
58
59 app = Dash(__name__)
60
61 app.layout = html.Div([
62
63     html.H1(
64         "IAPT + Prescribing Trends Dashboard",
65         style={
66             "textAlign": "center",
67             "marginBottom": "30px"
68         }
69     ),
70
71     html.Div([
72
73         html.Label("Select Region"),
74
75         dcc.Dropdown(
76             id='region-dropdown',
77             options=[
78                 {'label': r, 'value': r}
79                 for r in regions
80             ],
81             value=regions[0] if len(regions) > 0 else None,
82             clearable=False
83         )
84 ])
```



```
84
85 ], style={'width': '40%', 'margin': 'auto'}),
86
87 html.Br(),
88
89 dcc.Graph(id='iapt-chart'),
90
91 dcc.Graph(id='prescribing-chart')
92
93 ])
94
95 # -----
96 # Callback
97 # -----
98
99 @app.callback(
100     [
101         Output('iapt-chart', 'figure'),
102         Output('prescribing-chart', 'figure')
103     ],
104     [
105         Input('region-dropdown', 'value')
106     ]
107 )
108 def update_graphs(selected_region):
109
110     # IAPT chart
111     if not iapt_df.empty:
112         # -----
```

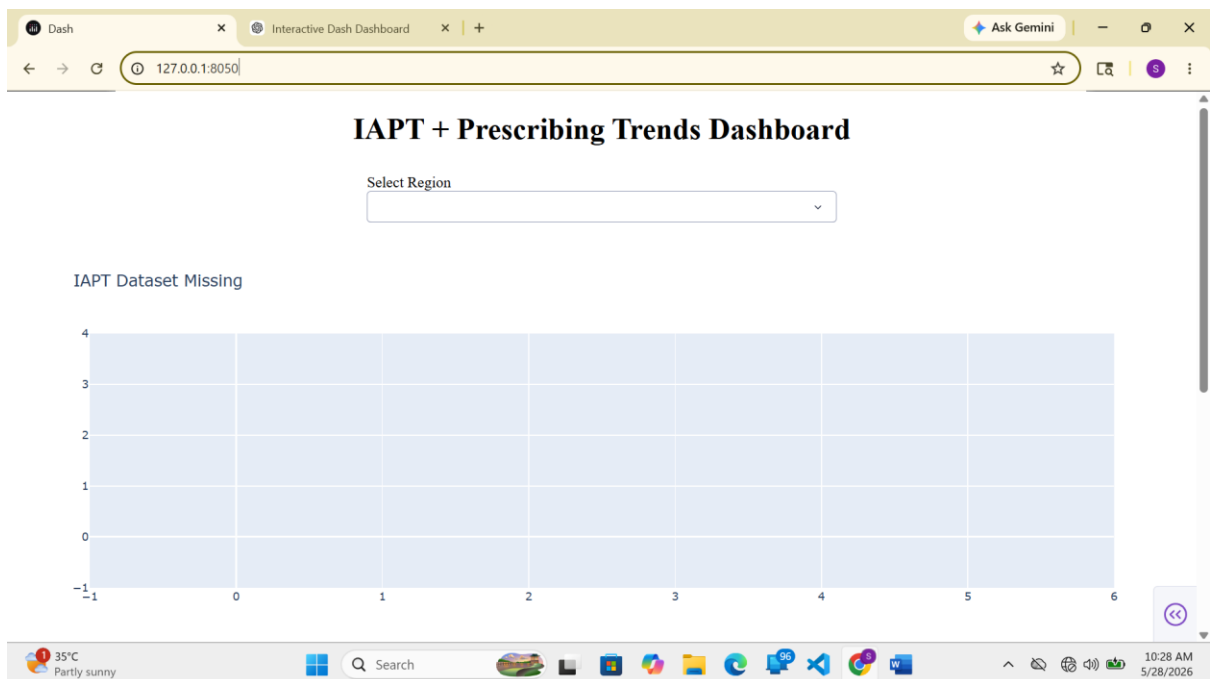
```
107 }
108 def update_graphs(selected_region):
109     # IAPT chart
110     if not iapt_df.empty:
111         filtered_iapt = iapt_df[
112             iapt_df[region_col] == selected_region
113         ]
114         fig1 = px.line(
115             filtered_iapt,
116             x=date_col,
117             y=metric_col,
118             title='IAPT Trends'
119         )
120     else:
121         fig1 = px.line(title='IAPT Dataset Missing')
122
123     # Prescribing chart
124     if not pres_df.empty:
125         pres_df.columns = pres_df.columns.str.strip()
126
127         numeric_cols = pres_df.select_dtypes(include='number').columns
128
129         if len(numeric_cols) > 0:
130             x_col = pres_df.columns[0]
```

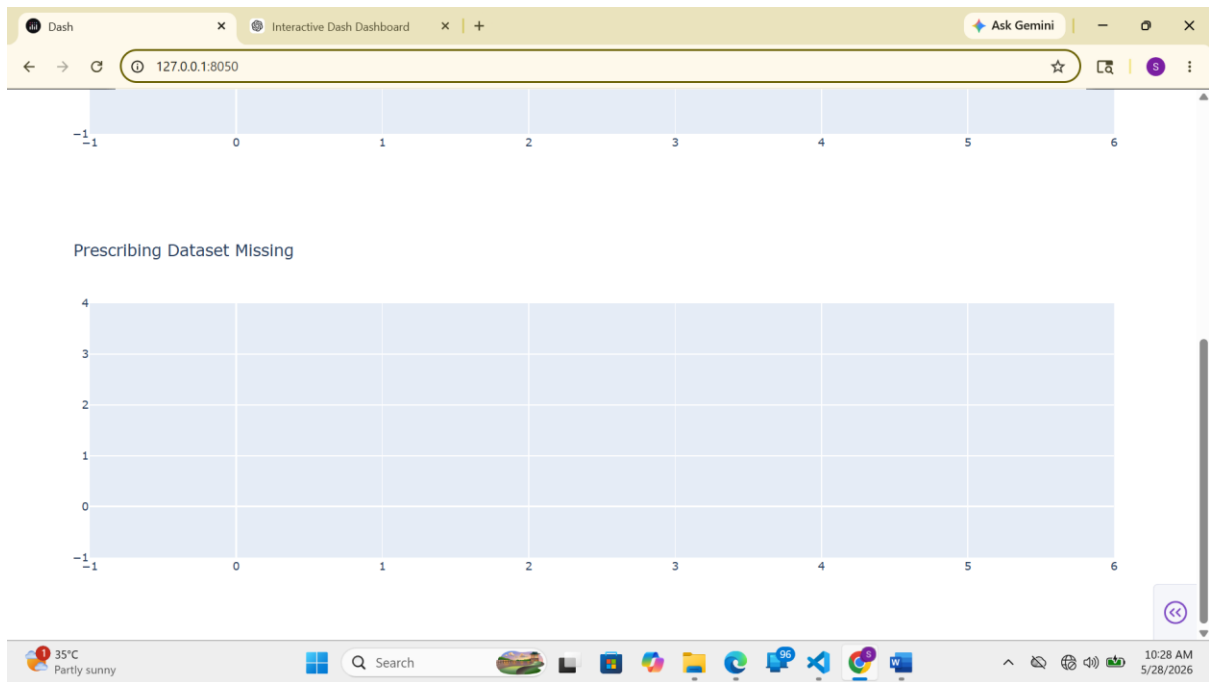
```
131         x_col = pres_df.columns[0]
132         y_col = numeric_cols[0]
133
134         fig2 = px.line(
135             pres_df,
136             x=x_col,
137             y=y_col,
138             title='Prescribing Trends'
139         )
140     else:
141         fig2 = px.line(title='No Numeric Columns')
142
143     else:
144         fig2 = px.line(title='Prescribing Dataset Missing')
145
146     return fig1, fig2
147
148 # -----
149 # Open browser automatically
150 # -----
151
152 def open_browser():
153     webbrowser.open_new("http://127.0.0.1:8050/")
154
155 # -----
156 # Run app
157 # -----
```



```
188 def update_graphs(selected_region):
189     fig2 = px.line(title='Prescribing Dataset Missing')
190     return fig1, fig2
191
192 # -----
193 # Open browser automatically
194 # -----
195
196 def open_browser():
197     webbrowser.open_new("http://127.0.0.1:8050/")
198
199 # -----
200 # Run app
201 # -----
202
203 if __name__ == "__main__":
204     Timer(1, open_browser).start()
205
206     app.run(
207         debug=True,
208         port=8050
209     )
```

Output





Conclusion

The Interactive Plotly Dash Dashboard successfully combines IAPT and prescribing datasets into a single analytical platform. Using Python, Pandas, Plotly, and Dash, the project provides an efficient way to explore mental health service trends and prescribing patterns.

The dashboard improves data visualization and user interaction through dropdown filters and dynamic charts. It enables users to monitor healthcare performance, identify regional differences, and analyze prescribing behavior effectively.

This project demonstrates how modern data visualization tools can transform raw healthcare datasets into meaningful insights for decision-making and reporting.